

Inverters



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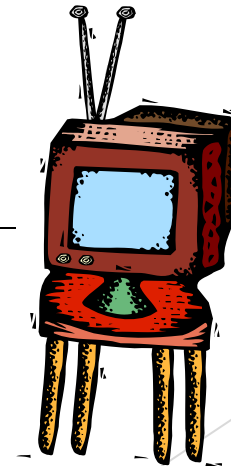
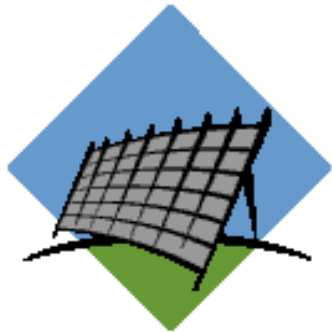
What is an Inverter?

- ▶ An inverter is an electrical circuit capable of turning DC power into AC power, while at the same time regulating the voltage, current, and frequency of the signal.



Why do I need an inverter?

- ▶ Most solar panels provide 12V DC power
- ▶ Household devices use 220V AC power
- ▶ Necessary for a grid tie in system

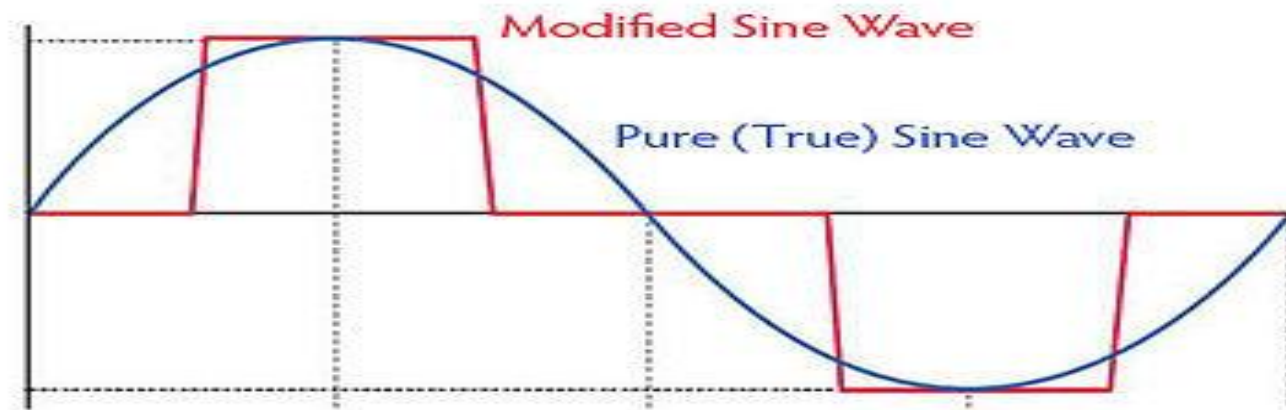
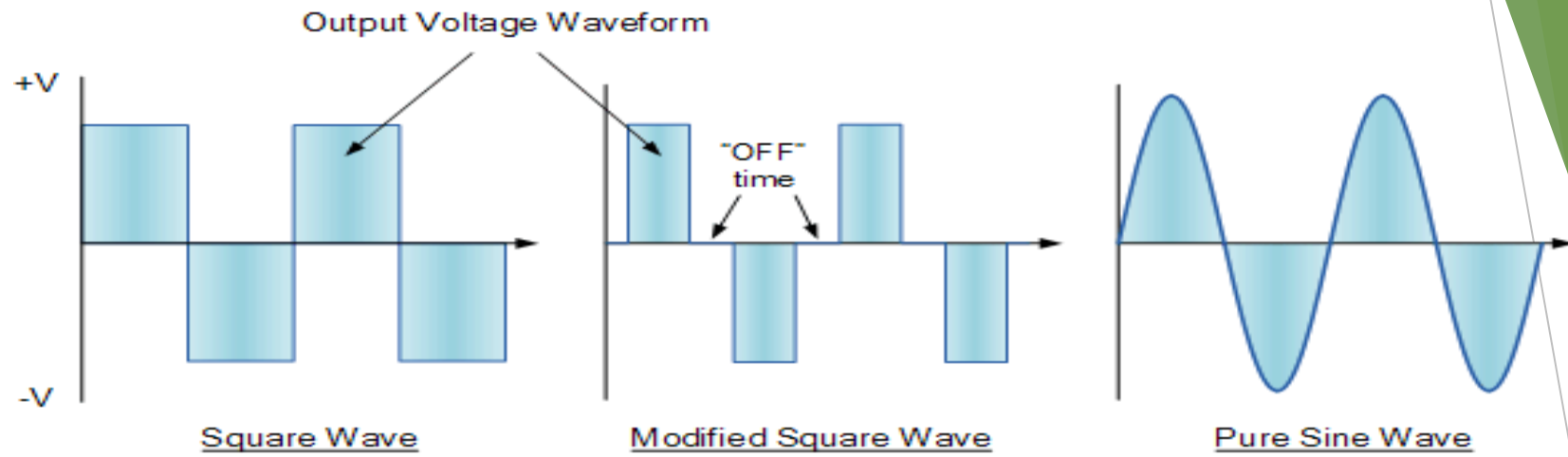


What kind of inverter do I want?

- ▶ Inverters come in all different shapes and sizes, for all different purposes
- ▶ Inverters vary in output from 50 - 5,000 W
- ▶ Several different methods of changing DC power to AC power
- ▶ Some inverters put out electricity of higher 'quality' than others. What does that mean?

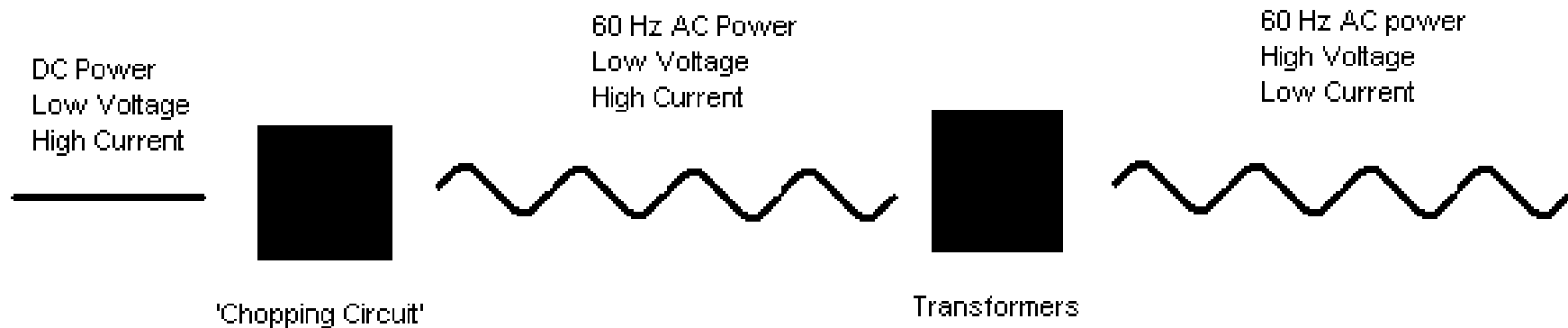
Different types of inverter output

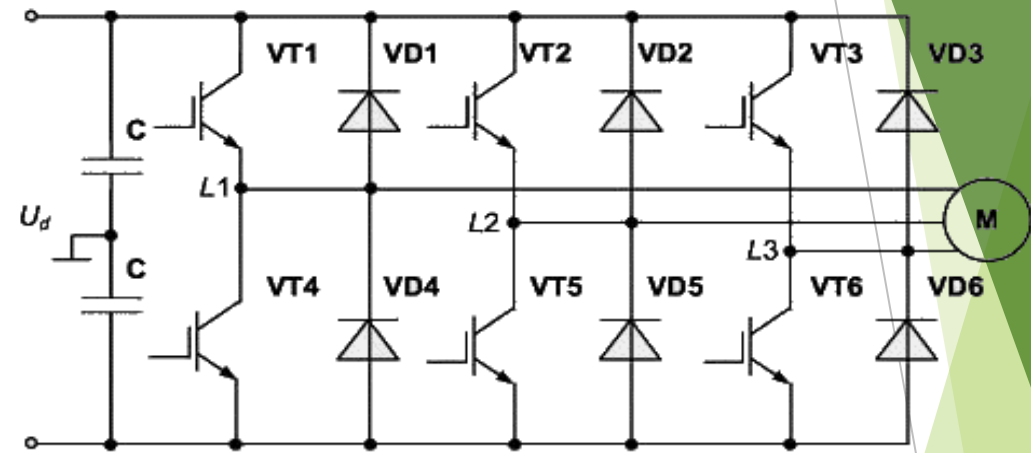
- ▶ Square wave inverters are basically obsolete
- ▶ Modified sine wave output will power almost everything in your house. However, things like power drills, or battery chargers may not work
- ▶ True sine wave output is exactly the same as the power provided by a utility company, and is necessary for a grid inter-tie system



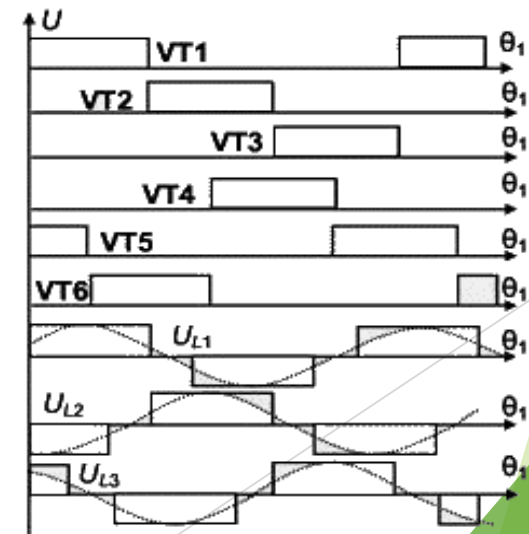
How do they work?

- ▶ Inverter takes low voltage high current power from a PV panel or battery and sends it through a 'chopping' circuit which changes it to low voltage high current AC power at 50Hz. The power then goes through a series of large transformers which output 220V AC power at 50Hz





a.



b.

Hybrid inverter

